

teries are quite costly, and this is where the gap is filled by iron-air batteries, which cost one-tenth of lithium batteries and whose iron electrodes have about 4000 cycles and air electrodes about 1000 life-time cycles.

Ups and downs

Advantages. Iron is the fourth most abundant element on Earth and inexpensive, which overcomes a significant problem with using lithium, a costly and rare element. The use of iron curtails the extensive use of

water in lithium mining and ground-water contamination.

Iron-air batteries can provide energy grids with dependable, safe, efficient, and longer-time energy storage capabilities than conventional technologies. While Li-ion batteries can provide about four hours of energy storage capacity, iron-air batteries can provide up to one hundred hours of storage. Therefore, iron-air batteries can function as a bridging technology during energy gaps, such as cloudy days, which would otherwise limit solar power plants.

When it comes to volumetric energy density, iron-air batteries perform better than Li-ion ones. At 9700Wh/l, they are about five times as high as today's Li-ion batteries (2000Wh/l). So, they are of particular interest for use in mobile appliances, where space requirements play an important role. Iron-air batteries are ten times as cheap as their Li-ion counterparts.

They contain no toxic elements, and the electrolyte is not flammable. So, they are quite safe.

Disadvantages. Iron-air batteries are large and recharge slowly. Their energy density is low and likely to stay that way, so there is no chance that this will become the favoured battery technology in the transport sector.

Challenges. Measures need to be taken to reduce the battery weight and increase electrode capacity, which is achievable through using a multi-cell bipolar stack with lightweight and cost-effective graphite bipolar plates. Electrode capacity can be increased by producing active mass of iron via electroplating, which is cost-effective. Using potash or caustic soda lye as an alkaline electrolyte should result in better coupling to the current collector. Moreover, the electrochemical deposition process can be used for targeted passivation of the electrode surface to prevent side reactions without impairing the actual function of the battery. This action would prevent the parasitic generation of hydrogen.

Market projection

Metal-air batteries are considered the most preferred replacement for lithium-based and other types of batteries, which are now widely used. The ability of metal-air batteries to use abundantly available and cheap metals like zinc, aluminium, and especially iron lowers manufacturing costs. The specific density and energy capacity of these batteries are higher than those of lithium-ion batteries.

The use of abundantly available metals as anodes and air as oxidants attracts manufacturers to produce these batteries on a commercial scale. Because of the lucrative opportunity to gain huge profits, manufacturers are now working with various

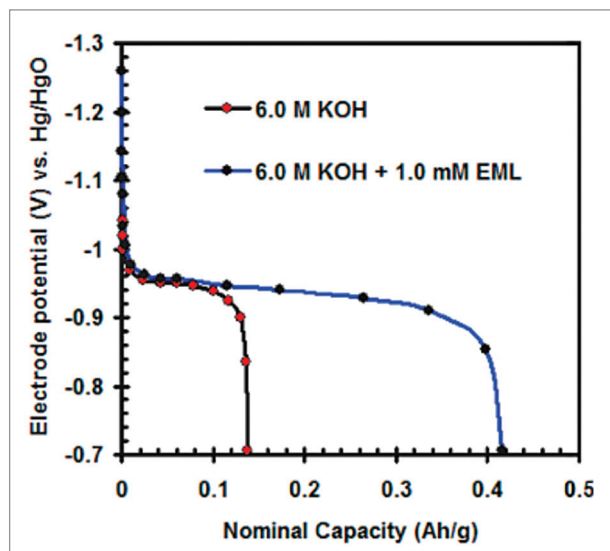


Fig. 13: Electrode potential vs nominal capacity during discharge of iron-air battery

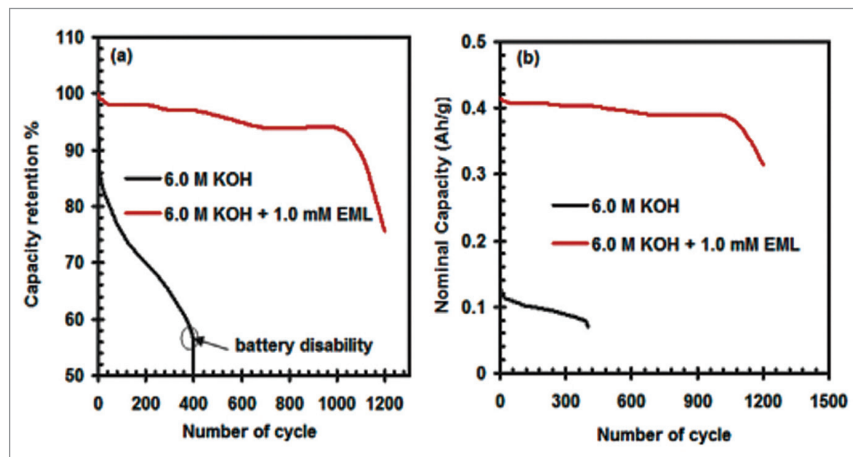


Fig. 14: (a) Capacity retention vs number of cycles, (b) Nominal capacity vs number of cycles of iron-air battery